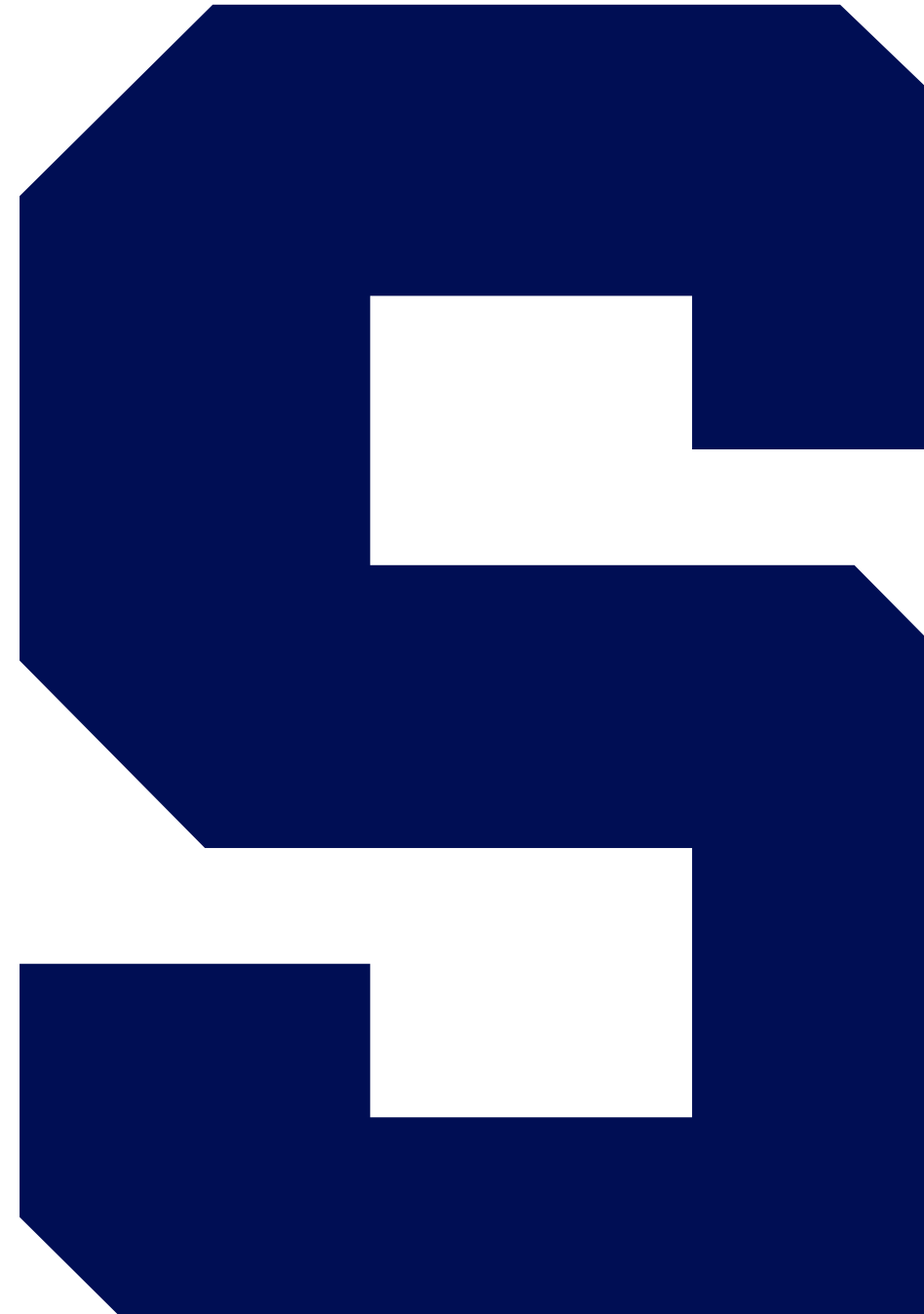


Definitions and Origins of Machine Learning



What is Machine Learning?

- **Tom M. Mitchell (1997) :**
 - "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ."
 - Source: *Machine Learning*, Tom M. Mitchell, McGraw Hill, 1997.
-

What is Machine Learning?

- **Arthur Samuel (1959) :**
 - "Machine Learning is the subfield of computer science that gives computers the ability to learn without being explicitly programmed."
 - Arthur Samuel is credited with coining the term "Machine Learning" in 1959. He's known for his work on the Samuel Checkers-playing Program.
-

What is Machine Learning?

- **Kevin P. Murphy (2012) :**
 - "Machine learning is a set of methods that can automatically detect patterns in data, and then use the uncovered patterns to predict future data, or to perform other kinds of decision-making under uncertainty (such as planning how to collect more data)."
 - Source: *Machine Learning: A Probabilistic Perspective* , Kevin P. Murphy, MIT Press, 2012.
-

What is Machine Learning?

- **Aurélien Géron (2015) :**
 - "Machine Learning is the science (and art) of programming computers so they can learn from data."
 - Source: *Hands-on Machine Learning* (your text-book!)
-

Machine Learning Origins

- **Early Theoretical Foundations :**

- “Reason... is nothing but Reckoning (that is, Adding and Subtracting) of the Consequences of general names agreed upon for the marking and signifying of our thoughts...” - *Thomas Hobbs, Leviathon (1651)*

- **Turing Test (1950) :**

- A "universal machine" that could simulate the behavior of any other machine.
- The "Turing Test" as a measure of a machine's ability to exhibit intelligent behavior indistinguishable from that of a human.

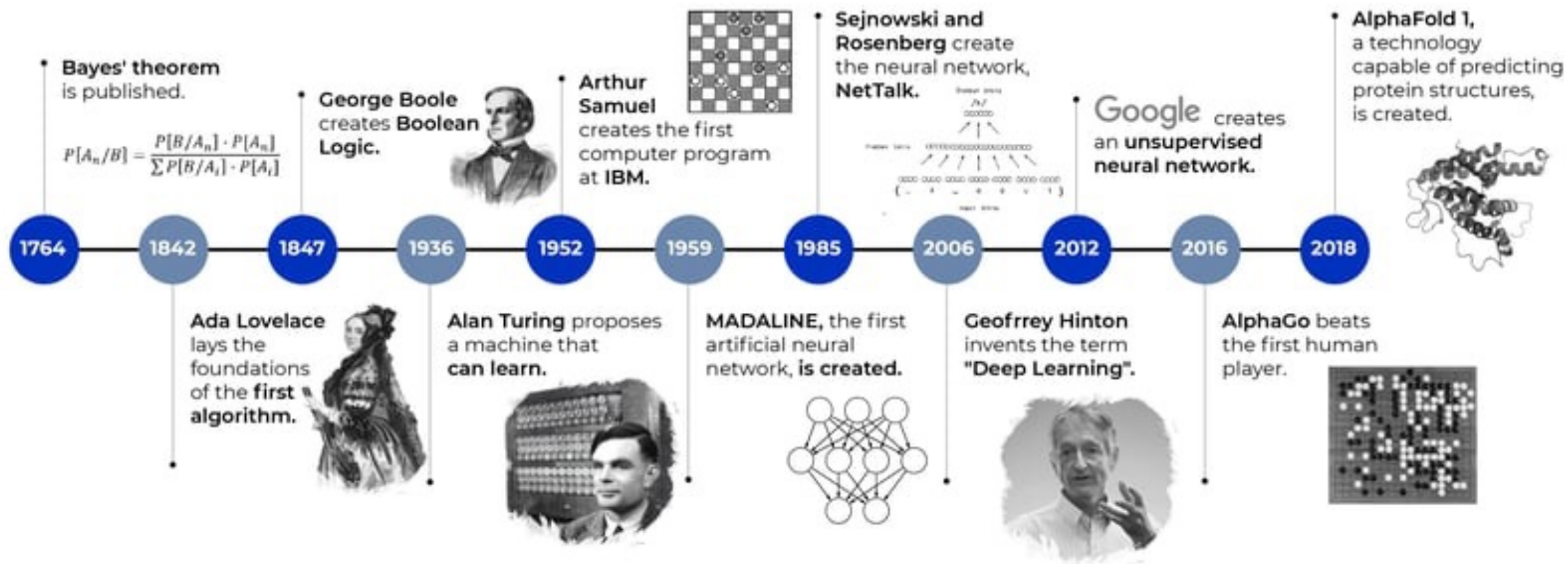
- **Advent of Electronic Computers (1940s-1950s) :**

- The creation of electronic computers provided the necessary tools for exploring computational approaches to problem-solving.

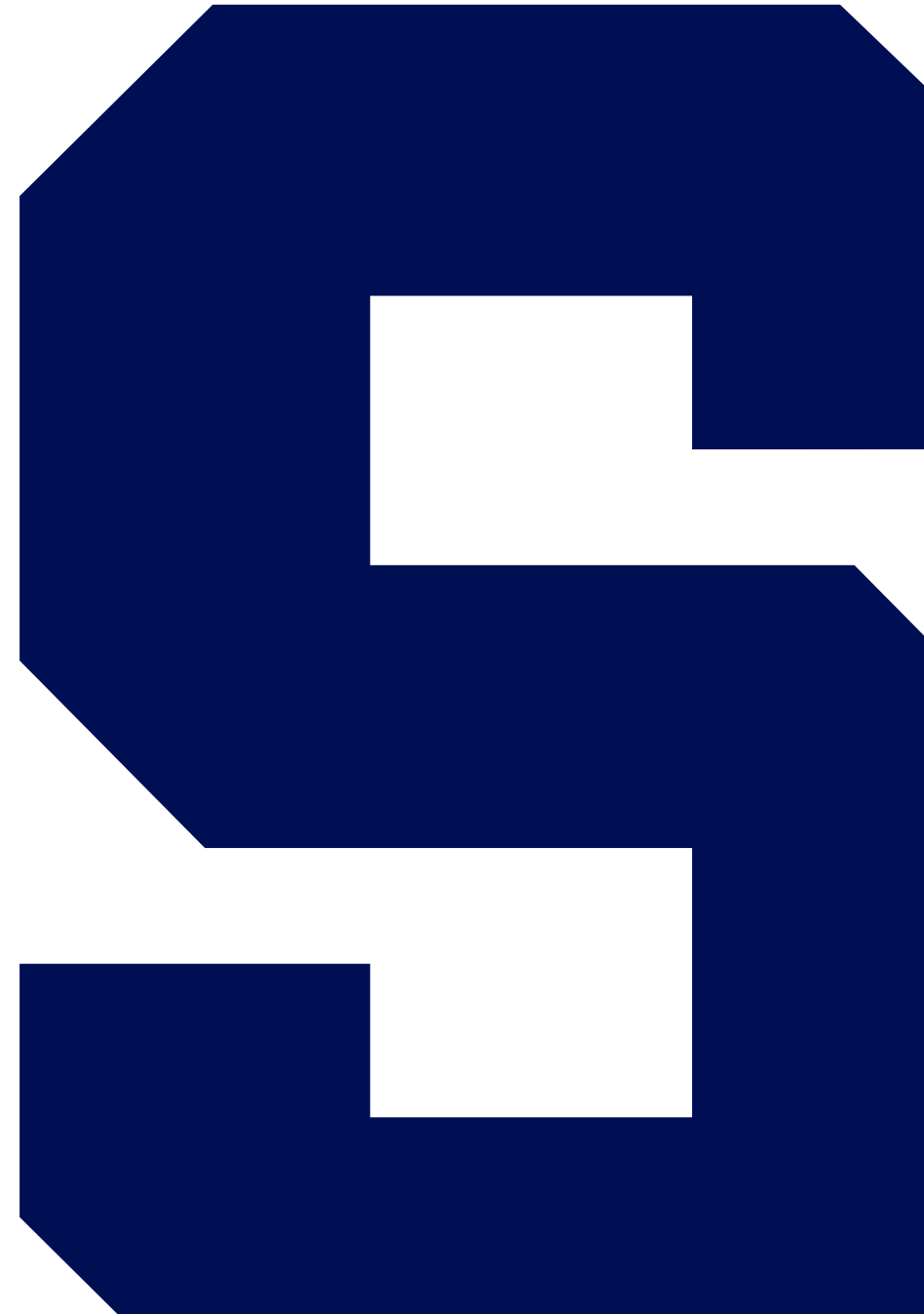
- **Birth of Artificial Intelligence (1956) :**

- At the Dartmouth Workshop in 1956, the term "Artificial Intelligence" (AI) was coined. This event is often considered the birth of AI as a formal academic discipline. Early AI research focused on problem-solving, logical reasoning, and symbolic methods.
-

MACHINE LEARNING TIMELINE



Applications of Machine Learning



Applications of Machine Learning

- **Healthcare & Life Sciences**

- Disease Identification & Diagnostics, Drug Discovery & Personalized Medicine, Predictive Analytics

- **Finance & Banking**

- Fraud Detection, Algorithmic Trading, Credit Scoring

- **E-Commerce & Retail**

- Recommendation Systems, Demand Forecasting, Customer Sentiment Analysis
-

Applications of Machine Learning

- **Automotive & Transportation**

- Autonomous Vehicles, Predictive Maintenance, Traffic Prediction

- **Energy & Utilities**

- Smart Grid Management, Predictive Maintenance for Equipment, Energy Consumption Forecasting

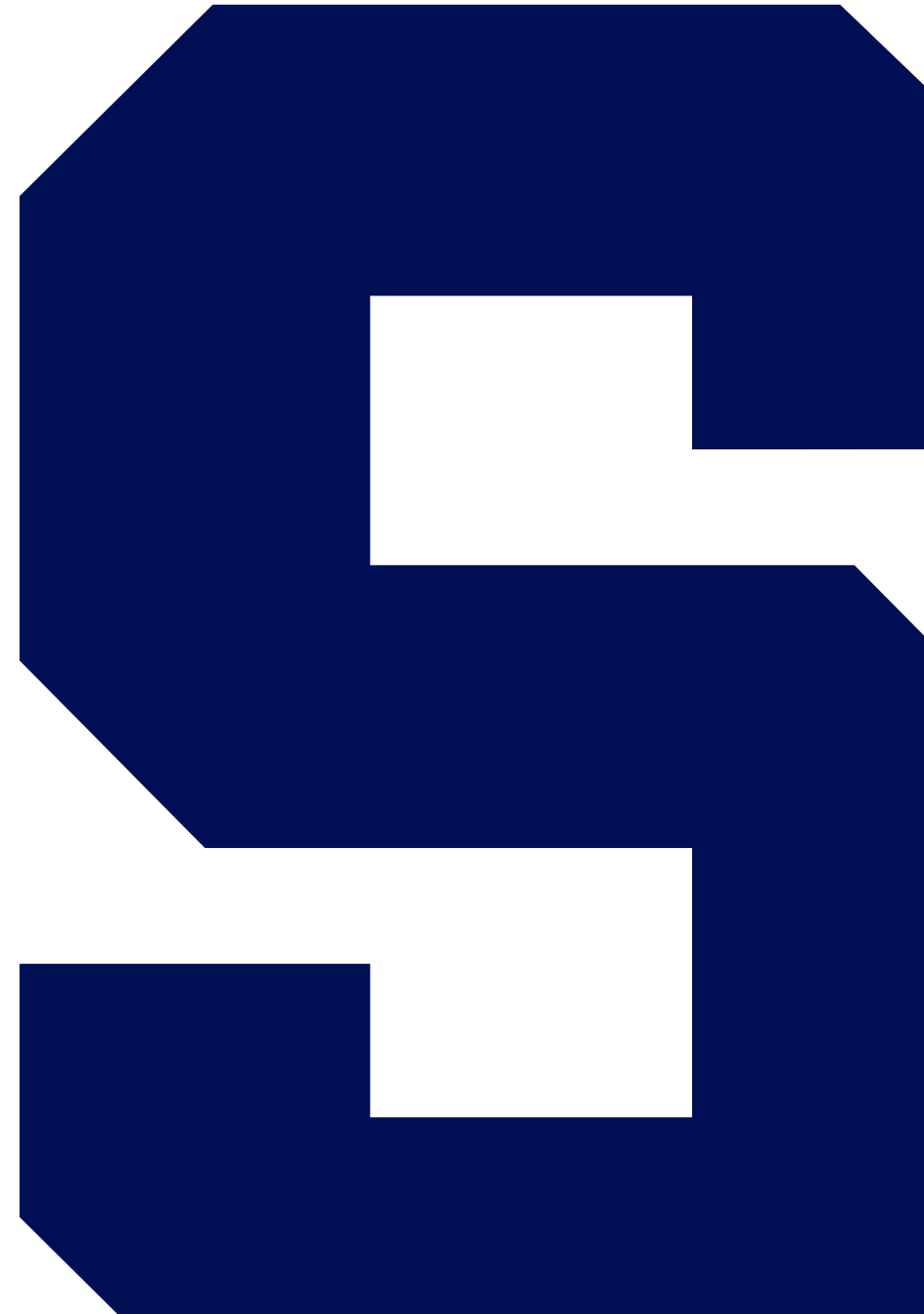
- **Entertainment & Media**

- Content Recommendation, Sentiment Analysis, Automated Content Creation
-

Activity

- What are the applications of machine learning that are most interesting to you personally?
 - What kinds of problems do you want to be able to solve using machine learning techniques?
-

Communication



The Importance of Communication

- **Translating Technicalities :**

- ML involves intricate algorithms, complex mathematical models, and vast amounts of data.
- Stakeholders are often more interested in the **actionable insights**.
- ML practitioners must articulate findings in a comprehensible and meaningful manner.

- **Understanding Stakeholder Needs:**

- Essential to understand the problem from a stakeholder's perspective. Aims?
Constraints?
 - Solutions must meet the needs of those who will use or be affected by the model.
-

The Importance of Communication

- **Setting Clear Expectations :**

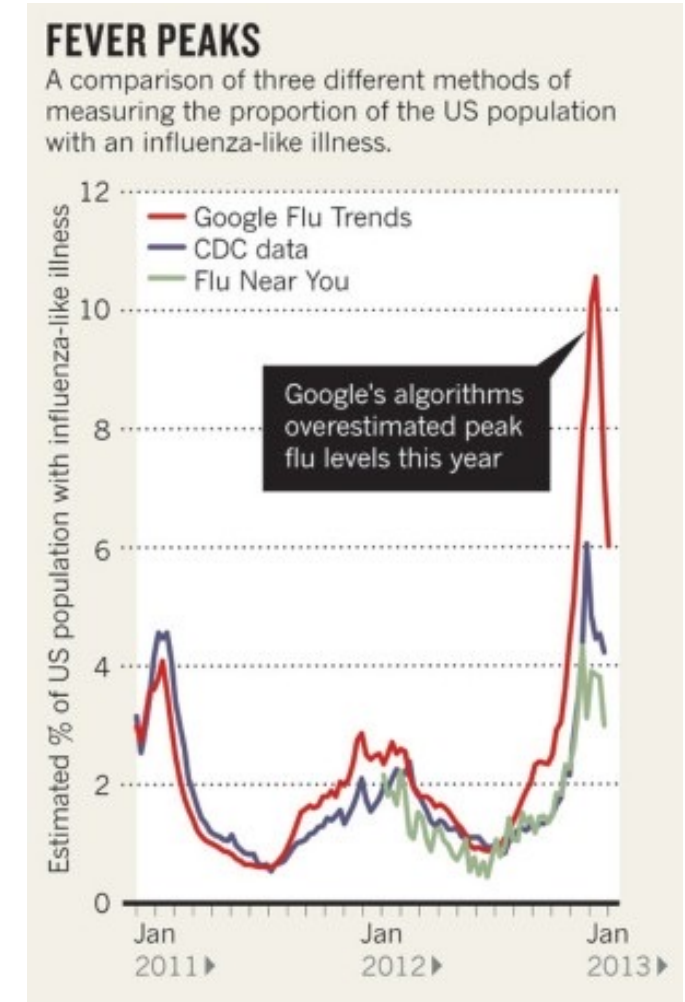
- Overhype leads to disappointment and mistrust.
- Misuse can cause substantive harm.
- Vital to communicate what a model can and cannot do, its limitations, and any uncertainties involved.

- **Ethical Considerations :**

- Increasing concerns about ML's ethical implications, potential biases, and fairness.
 - Transparent communication about how models work, and their potential pitfalls is crucial.
-

Failures of Communication

- **Google Flu Trends :**
 - Google's attempt to predict flu outbreaks using search query data.
 - High expectations and inconsistent performance led to skepticism about the utility of big data.
 - Effort terminated.



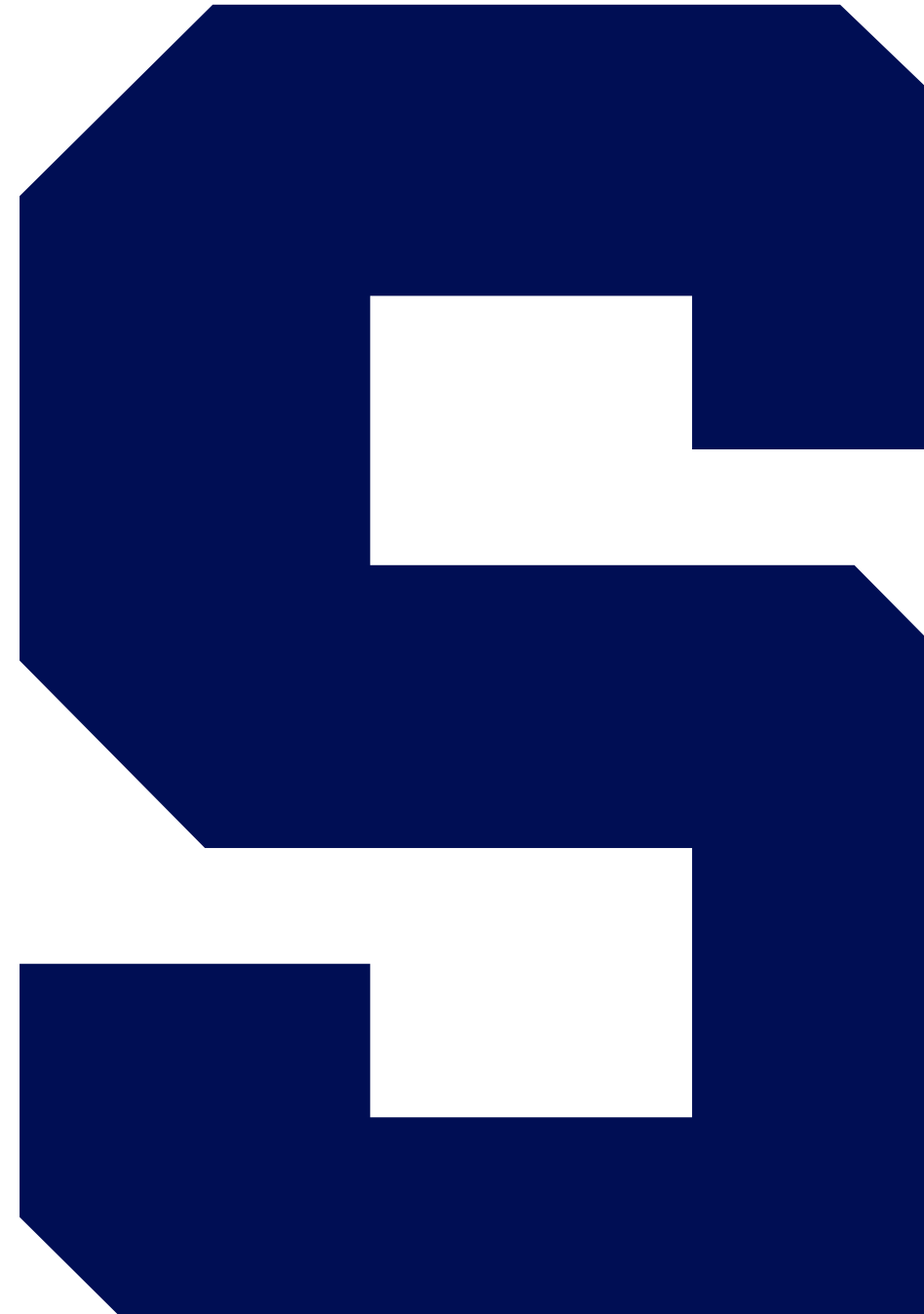
Source: <https://www.nature.com/articles/494155a>

Failures of Communication

- **IBM Watson for Oncology :**
 - Promised to revolutionize cancer treatment by providing personalized treatment recommendations.
 - Erroneous treatment recommendations & over-promotion of Watson's capabilities without clear communication about its limitations.
 - Effort terminated.



The Python Machine Learning Ecosystem



Introduction to the Python ML Ecosystem

- Machine Learning in Python is characterized by an ecosystem of **libraries**, each specializing in different aspects of the ML workflow. This ecosystem simplifies the process of developing and implementing ML models and ensures efficiency and versatility.
 - The **integration** between these libraries allows for a streamlined workflow, from initial data handling to final model deployment.
 - There are many **online resources** for learning to use these libraries, and chatbots like ChatGPT are generally quite fluent with this ecosystem.
-

Python

Python is a high-level programming language with a straightforward syntax that helps programmers develop highly readable code.

Python is the foundation upon which the following libraries are built, offering a unified scripting environment.

Resources:

- **Official Python Documentation:** docs.python.org
- **Python Tutorial by Python.org:** <https://docs.python.org/3/tutorial/>
- **Real Python:** Offers tutorials and articles on various Python topics - <https://realpython.com/>



Pandas



Pandas is typically the starting point in the data science workflow. Provides the DataFrame class for efficient storage and manipulation of tabular data, and incorporates tools for handling missing data, merging, reshaping, and slicing datasets. It's used for:

- **Data Import and Wrangling:** Reading data from various sources and performing initial transformations.
- **Data Analysis:** Simplifies exploratory data analysis with built-in functions.
- **Interfacing with Other Libraries:** Pandas integrates well with libraries like NumPy for numerical operations, Matplotlib and Seaborn for visualization, and can feed data directly into machine learning models in Scikit-learn.

Resources:

- **Official Pandas Documentation:** <https://pandas.pydata.org/docs/>
 - **Pandas Getting Started Tutorials:** https://pandas.pydata.org/docs/getting_started/index.html#getting-started
 - **Kaggle Pandas Course:** <https://www.kaggle.com/learn/pandas>
-



NumPy excels in numerical computations. It provides efficient operations with arrays and matrices, thanks to its implementation in C. NumPy supports a wide range of mathematical operations, which is fundamental in ML for data manipulation and transformation.

- **A Backbone for Pandas and Scikit-Learn:** Many Pandas operations are built on NumPy, and Scikit-learn utilizes NumPy arrays for optimal performance in model training and predictions.
- **Essential for Numerical Analysis:** Any heavy numerical computation, especially involving arrays and matrices, relies on NumPy for efficiency.

Resources:

- **Official NumPy Documentation:** <https://numpy.org/doc/stable/>
 - **NumPy User Guide:** <https://numpy.org/doc/stable/user/index.html>
 - **SciPy Lectures on NumPy:** <http://scipy-lectures.org/intro/numpy/index.html>
-

Matplotlib and Seaborn



Visualization is essential for understanding data distributions, patterns, and insights, and hence indispensable in ML for data exploration and analysis of results. These libraries are the most widely used visualization tools in the ML ecosystem, and directly use Pandas data structures for plotting; NumPy's numerical capabilities augment plotting functionalities.

- **Matplotlib:** A versatile library for creating a wide range of static, animated, and interactive plots.
- **Seaborn:** Built on top of Matplotlib, it provides a high-level interface for drawing attractive and informative statistical graphics, making complex visualizations more accessible.

Resources:

- **Official Documentation:** <https://matplotlib.org/stable/contents.html>; <https://seaborn.pydata.org/>
 - **Tutorials:** <https://matplotlib.org/stable/tutorials/index.html>; <https://seaborn.pydata.org/tutorial.html>
 - **Python Graph Gallery:** <https://www.python-graph-gallery.com/matplotlib>
 - **DataCamp Seaborn Tutorial:** <https://www.datacamp.com/community/tutorials/seaborn-python-tutorial>
-

Scikit-Learn

Scikit-Learn is where the ML models come to life. It provides a wide array of supervised and unsupervised learning algorithms, as well as tools for feature selection, normalization, and cross-validation. While Scikit-Learn can use Pandas DataFrames directly, it is optimized for NumPy arrays. It is therefore important to understand data conversion and manipulation across these libraries.

Resources:

- **Official Scikit-Learn Documentation:** <https://scikit-learn.org/stable/documentation.html>
- **Scikit-Learn User Guide:** https://scikit-learn.org/stable/user_guide.html
- **Scikit-Learn Tutorials:** <https://scikit-learn.org/stable/tutorial/index.html>



Python ML Ecosystem Wrap-up

In this ecosystem, each library has a distinct role but is also intricately **connected** with the others:

Pandas and NumPy handle the **data preparation**; Matplotlib and Seaborn assist in **data visualization**; and Scikit-Learn is used for building and evaluating **machine learning models**.

Understanding how to navigate these libraries and leverage their interconnectedness is crucial for any aspiring data scientist or ML practitioner.
